

FN585 Financial Modelling & Dealing

Week 3 seminar

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First difference

- First difference: $\Delta Y_t = Y_t - Y_{t-1}$
- Log difference: $\Delta \ln Y_t = \ln Y_t - \ln Y_{t-1}$

The percentage change of Y_t between periods $t - 1$ and t is approximately $100\Delta \log Y_t$, where this approximation is most accurate when the percentage change is small.

Autoregressive Forecasts

- How to predict the future? Start with the immediate past!
- Autoregression for Y_t models $E(Y_t | Y_{t-1}, Y_{t-2}, \dots)$

That is: an autoregression models the conditional mean of Y_t as a function of its own lagged values

- An (estimated) autoregressive model; for Y_t allows you to forecast Y_{t+1} from data on $Y_t, Y_{t-1}, \dots, Y_{t-p}$

1ST-ORDER AUTOREGRESSIONS

The 1st-order autoregressive model is

$$E(Y_t | Y_{t-1}, Y_{t-2}, \dots) = E(Y_t | Y_{t-1}) = \beta_0 + \beta_1 Y_{t-1},$$

where β_0 and β_1 are unknown (population) coefficients.

The AR(1)-model is

$$Y_t = \beta_0 + \beta_1 Y_{t-1} + u_t$$

where $u_t = Y_t - E(Y_t | Y_{t-1})$ is an error term with

$$E(u_t | Y_{t-1}, Y_{t-2}, \dots) = E(u_t | Y_{t-1}) = 0$$

Estimating the AR(1) Model

- The AR(1)-model is a simple regression of Y_t (the dependent variable) on Y_{t-1} (the regressor)
- Estimate β_0 and β_1 by OLS using $(Y_t; t = 1, 2, \dots, T)$
- Various ways of doing that in R, including using the `lm()` function we have used thus far

1ST-ORDER AUTOREGRESSIONS

```
# load US macroeconomic data
usmac_qt <- read_xlsx("us_macro_quarterly.xlsx",
                     sheet = 1,col_types = c("text", rep("numeric", 9)))

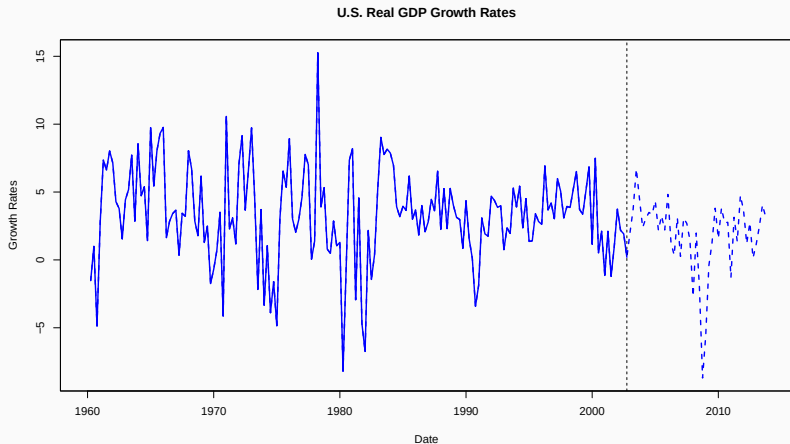
## New names:
## * ' ' -> '...1'

# Fix format of date
usmac_qt$...1 <- as.yearqtr(usmac_qt$...1, format = "%Y:0%q")
# Relabel column names in dataframe
colnames(usmac_qt) <- c("Date", "GDPC96", "JAPAN_IP", "PCEPTI", "GS10",
                       "GS1", "TB3MS", "UNRATE", "EXUSUK", "CPIAUCSL")

# GDP series as xts object
GDP <- xts(usmac_qt$GDPC96, usmac_qt$Date)["1960::2013"]
# GDP growth series as xts object
GDPGrowth <- xts(400 * log(GDP/lag(GDP)))
# Training data
GDPGrowth.train <- na.omit(GDPGrowth["1960::2002"])
```

US GDP GROWTH

```
plot(as.zoo(GDPGrowth), col = "blue", lwd = 2, lty = "dashed",  
     ylab = "Growth Rates", xlab = "Date", main = "U.S. Real GDP Growth Rates")  
abline(v = 2002.75, lty = "dashed")  
lines(as.zoo(GDPGrowth.train), lwd = 2, col = "blue")
```



AR(1)-MODEL BY AR.OLS()

```
# AR(1) mode by ar.ols-function
ar1gdp.arols <- ar.ols(GDPGrowth.train, order.max = 1,
                      demean = FALSE, intercept = TRUE)
ar1gdp.arols # Print output to console

##
## Call:
## ar.ols(x = GDPGrowth.train, order.max = 1, demean = FALSE)
##
## Coefficients:
##      1
## 0.3017
##
## Intercept: 2.349 (0.3507)
##
```

AR(1)-MODEL BY dynlm()

```
# AR(1) model by lm-function
```

```
library(dynlm)
```

```
ar1gdp.lm <- dynlm(ts(GDPGrowth.train) ~ L(ts(GDPGrowth.train)))
```

```
ar1gdp.lm # Print to console
```

```
##
```

```
## Time series regression with "ts" data:
```

```
## Start = 2, End = 171
```

```
##
```

```
## Call:
```

```
## dynlm(formula = ts(GDPGrowth.train) ~ L(ts(GDPGrowth.train)))
```

```
##
```

```
## Coefficients:
```

```
##              (Intercept)    L(ts(GDPGrowth.train))
```

```
##              2.3492              0.3017
```

dynlm()-stargazer

```
library(stargazer)
```

```
# Summary
```

```
stargazer(ar1gdp.lm, type="text")
```

```
##
## =====
##                               Dependent variable:
##                               -----
##                               ts(GDPGrowth.train)
## -----
## L(ts(GDPGrowth.train))          0.302***
##                               (0.073)
##
## Constant                        2.349***
##                               (0.353)
##
## -----
## Observations                    170
## R2                             0.092
## Adjusted R2                     0.086
## Residual Std. Error             3.303 (df = 168)
## F Statistic                     16.955*** (df = 1; 168)
## =====
## Note:                          *p<0.1; **p<0.05; ***p<0.01
```

AR(1) Model Regression Results for GDP Growth

	Dependent variable:
	$GDPGrowth_t$
$GDPGrowth_{t-1}$	0.302*** (0.073)
Constant	2.349*** (0.353)
Observations	170
R^2	0.092
Adjusted R^2	0.086
Residual Std. Error	3.303
F Statistic	16.955***
Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$	

Table 1: AR(1) Model Regression Results for GDP Growth

This table presents the results from an autoregressive AR(1) model estimated using ordinary least squares (OLS), where GDP growth in period t ($GDPGrowth_t$) is regressed on its lagged value ($GDPGrowth_{t-1}$). The coefficient on $GDPGrowth_{t-1}$ is 0.302, statistically significant at the 1% level ($p < 0.01$), indicating a moderate degree of persistence in GDP growth. The constant term is also positive and significant. The model explains approximately 9.2% of the variation in GDP growth, as indicated by $R^2 = 0.092$, with an adjusted R^2 of 0.086. The F-statistic (16.955) suggests that the overall model is statistically significant. Standard errors (reported in parentheses) are non-robust and may not account for heteroskedasticity or autocorrelation.

ROBUST STANDARD ERRORS

```
# Output w/ robust standard errors  
library(sandwich)  
coeftest(ar1gdp.lm, vcov. = vcovHAC)
```

```
##  
## t test of coefficients:  
##  
##               Estimate Std. Error t value Pr(>|t|)  
## (Intercept)      2.349176   0.393934   5.9634 1.418e-08 ***  
## L(ts(GDPGrowth.train)) 0.301744   0.077997   3.8686 0.0001563 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

ROBUST STANDARD ERRORS

```
# Output w/ robust standard errors
```

```
ar1gdp.Roblm<-coeftest(ar1gdp.lm, vcov. = vcovHAC)
```

```
stargazer(ar1gdp.lm, ar1gdp.Roblm, type="text")
```

```
##
## =====
##                               Dependent variable:
##                               -----
##                               ts(GDPGrowth.train)
##                               dynamic      coefficient
##                               linear       test
##                               (1)         (2)
## -----
## L(ts(GDPGrowth.train))      0.302***      0.302***
##                               (0.073)      (0.078)
##
## Constant                    2.349***      2.349***
##                               (0.353)      (0.394)
##
## -----
## Observations                170
## R2                          0.092
## Adjusted R2                 0.086
## Residual Std. Error        3.303 (df = 168)
## F Statistic                 16.955*** (df = 1; 168)
## =====
## Note:                       *p<0.1; **p<0.05; ***p<0.01
```

AR(1) Model Regression Results for GDP Growth

Table 2: AR(1) Model Regression Results for GDP Growth

<i>Dependent variable:</i> <i>GDPGrowth_t</i>		
	<i>Model 1</i>	<i>Model 2</i>
<i>GDPGrowth_{t-1}</i>	0.302*** (0.073)	0.302*** (0.078)
Constant	2.349*** (0.353)	2.349*** (0.394)
Observations	170	
R ²	0.092	
Adjusted R ²	0.086	
Residual Std. Error	3.303	
F Statistic	16.955***	

Note: Model 1 uses non-robust standard errors; Model 2 uses HAC standard errors.

*p<0.1; **p<0.05; ***p<0.01

One-Step Ahead US GDP Growth Forecast

$$Y_t = 2.35 + 0.30Y_{t-1} + \hat{u}_t$$

(0.39) (0.08)

- The one-step ahead forecast for US GDP growth is modeled as:

$$\hat{Y}_{T+1|T} = 2.35 + 0.30Y_T$$

(0.39) (0.08)

- where the numbers in parentheses are the standard errors:
- The forecast error is:

$$Y_{t+1} - \hat{Y}_{t+1|t}$$

ONE-STEP AHEAD US GDP GROWTH FORECAST

```
# Produce one-step ahead forecast of GDP growth for 2003:Q1
K=length(GDPGrowth.train)
fcast.ar1.byhand <- ar1gdp.lm$coefficients %*% c(1, GDPGrowth.train[K])
print(paste("the forecast:",fcast.ar1.byhand)) # Print to console
```



```
## [1] "the forecast: 2.40737477465024"
```



```
# Forecast error
fcast.ar1.error <- as.numeric(GDPGrowth["2003"][1]) - fcast.ar1.byhand
print(paste("the forecast error:",fcast.ar1.error)) # Print to console
```

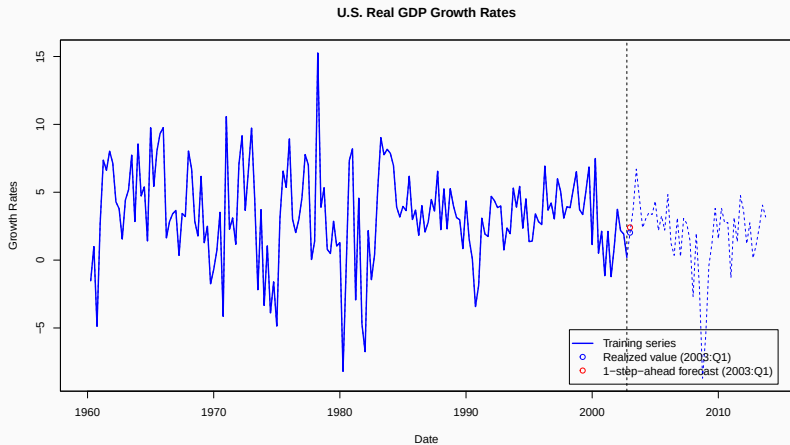


```
## [1] "the forecast error: -0.385880152881912"
```

US GDP GROWTH AND 1-STEP AHEAD AR(1)-FORECAST

```
# Plot US quarterly GDP time series w/ forecast
plot(as.zoo(GDPGrowth), col = "blue",lwd = 1,
     lty = "dashed",
     ylab = "Growth Rates", xlab = "Date",
     main = "U.S. Real GDP Growth Rates")
abline(v = 2002.75, lty = "dashed")
lines(as.zoo(GDPGrowth.train), lwd = 2, col = "blue")
points(2003,fcast.ar1.byhand, cex = 1, col ="red")
points(2003,GDPGrowth["2003"][1], cex = 1, col ="blue")
legend("bottomright", inset = 0.02, c("Training series",
   "Realized value (2003:Q1)",
   "1-step-ahead forecast (2003:Q1)"),
col=c("blue","blue","red"),
pch=c(NA,1,1),lty=c(1,NA,NA),lwd = c(2,NA,NA))
```

US GDP GROWTH AND 1-STEP AHEAD AR(1)-FORECAST



The p -th Order Autoregressive Model

- The order- p autoregressive model is:

$$E(Y_t | Y_{t-1}, \dots) = E(Y_t | Y_{t-1}, \dots, Y_{t-p}) = \beta_0 + \sum_{k=1}^p \beta_k Y_{t-k}$$

where $\beta_0, \beta_1, \dots, \beta_p$ are unknown (population) coefficients

- Add error $u_t = Y_t - E(Y_t | Y_{t-1}, \dots, Y_{t-p})$ to get $AR(p)$ -model:

$$Y_t = \beta_0 + \sum_{k=1}^p \beta_k Y_{t-k} + u_t$$

Estimating the AR(p)-Model

- The AR(p)-model is a multiple regression of Y_t (dependent variable) on $Y_{t-1}, \dots, Y_{t-p}$ (regressors)
- Estimate $\beta_0, \beta_1, \dots, \beta_p$ by OLS using $(Y_t; t = 1, 2, \dots, T)$
- Various ways of doing that in R, including using the `lm()` function we have used thus far

AR(4)-MODEL FOR US GDP GROWTH

```
library(dynlm) # Load dynlm to estimate higher-order autoregressions
ar4gdp.dynlm <- dynlm(ts(GDPGrowth.train) ~ L(ts(GDPGrowth.train)) +
                     L(ts(GDPGrowth.train), 2) +
                     L(ts(GDPGrowth.train), 3) +
                     L(ts(GDPGrowth.train), 4))
ar4gdp.dynlm # Print to console
```

```
##
## Time series regression with "ts" data:
## Start = 5, End = 171
##
## Call:
## dynlm(formula = ts(GDPGrowth.train) ~ L(ts(GDPGrowth.train)) +
##       L(ts(GDPGrowth.train), 2) + L(ts(GDPGrowth.train), 3) + L(ts(GDPGrowth.train),
##       4))
##
## Coefficients:
##              (Intercept)              L(ts(GDPGrowth.train))
##              2.05005              0.25849
## L(ts(GDPGrowth.train), 2) L(ts(GDPGrowth.train), 3)
##              0.15456              -0.03891
## L(ts(GDPGrowth.train), 4)
##              0.02788
```

AR(4)-MODEL FOR US GDP GROWTH

Summary

```
stargazer(ar4gdp.dynlm, type="text")
```

```
##
## =====
##                               Dependent variable:
##                               -----
##                               ts(GDPGrowth.train)
## -----
## L(ts(GDPGrowth.train))          0.258***
##                               (0.079)
##
## L(ts(GDPGrowth.train), 2)       0.155*
##                               (0.080)
##
## L(ts(GDPGrowth.train), 3)      -0.039
##                               (0.080)
##
## L(ts(GDPGrowth.train), 4)       0.028
##                               (0.077)
##
## Constant                       2.050***
##                               (0.458)
##
## -----
## Observations                    167
## R2                             0.111
## Adjusted R2                    0.090
## Residual Std. Error            3.265 (df = 162)
```

AR(4)-MODEL FOR US GDP GROWTH

```
# Output w/ robust standard errors
```

```
library(sandwich)
```

```
coeftest(ar4gdp.dynlm, vcov. = vcovHAC)
```

```
##
```

```
## t test of coefficients:
```

```
##
```

```
##
```

	Estimate	Std. Error	t value	Pr(> t)	
--	----------	------------	---------	----------	--

## (Intercept)	2.050052	0.553998	3.7005	0.0002945	***
----------------	----------	----------	--------	-----------	-----

## L(ts(GDPGrowth.train))	0.258488	0.084864	3.0459	0.0027088	**
---------------------------	----------	----------	--------	-----------	----

## L(ts(GDPGrowth.train), 2)	0.154563	0.102419	1.5091	0.1332157	
------------------------------	----------	----------	--------	-----------	--

## L(ts(GDPGrowth.train), 3)	-0.038910	0.076938	-0.5057	0.6137312	
------------------------------	-----------	----------	---------	-----------	--

## L(ts(GDPGrowth.train), 4)	0.027879	0.099559	0.2800	0.7798122	
------------------------------	----------	----------	--------	-----------	--

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ar4gdp.robdynlm<-coeftest(ar4gdp.dynlm, vcov. = vcovHAC)
```


AR(4)-MODEL FOR US GDP GROWTH

Output w/ robust standard errors

```
stargazer(ar4gdp.dynlm,ar4gdp.robdynlm, type="text")
```

```
##
## =====
##                               Dependent variable:
##                               -----
##                               ts(GDPGrowth.train)
##                               dynamic      coefficient
##                               linear        test
##                               (1)          (2)
## -----
## L(ts(GDPGrowth.train))          0.258***      0.258***
##                               (0.079)      (0.085)
##
## L(ts(GDPGrowth.train), 2)       0.155*        0.155
##                               (0.080)      (0.102)
##
## L(ts(GDPGrowth.train), 3)      -0.039        -0.039
##                               (0.080)      (0.077)
##
## L(ts(GDPGrowth.train), 4)       0.028        0.028
##                               (0.077)      (0.100)
##
## Constant                       2.050***      2.050***
##                               (0.458)      (0.554)
##
## -----
## Observations                     167
```

AR(4)-MODEL FOR US GDP GROWTH

Dependent variable: $GDPGrowth_t$		
	Model 1	Model 2
$GDPGrowth_{t-1}$	0.258*** (0.079)	0.258*** (0.085)
$GDPGrowth_{t-2}$	0.155* (0.080)	0.155 (0.102)
$GDPGrowth_{t-3}$	-0.039 (0.080)	-0.039 (0.077)
$GDPGrowth_{t-4}$	0.028 (0.077)	0.028 (0.100)
Constant	2.050*** (0.458)	2.050*** (0.554)
Observations	167	
R ²	0.111	
Adjusted R ²	0.090	
Residual Std. Error	3.265	
F Statistic	5.080***	

Note:

Model 1 uses non-robust standard errors; Model 2 uses HAC standard errors.

*p<0.1; **p<0.05; ***p<0.01

One-Step Ahead US GDP Growth Forecast

$$Y_t = 2.05 + 0.26Y_{t-1} + 0.15Y_{t-2} + 0.04Y_{t-3} + 0.03Y_{t-4} + \hat{u}_t$$

(0.55) (0.09) (0.10) (0.08) (0.10)

- The one-step ahead forecast for US GDP growth is modeled as:

$$\hat{Y}_{T+1|T} = 2.05 + 0.26Y_T + 0.15Y_{T-1} + 0.04Y_{T-2} + 0.03Y_{T-3} + \hat{u}_T$$

(0.55) (0.09) (0.10) (0.08) (0.10)

- where the numbers in parentheses are the standard errors:
- The forecast error is:

$$Y_{T+1} - \hat{Y}_{T+1|T}$$

ONE-STEP AHEAD US GDP GROWTH FORECAST

```
# Produce one-step ahead forecast of GDP growth for 2003:Q1
K=length(GDPGrowth.train)
fcast.ar4 <- ar4gdp.dynlm$coefficients %*% c(1, GDPGrowth.train[K],
                                             GDPGrowth.train[(K-1)],
                                             GDPGrowth.train[(K-2)],
                                             GDPGrowth.train[(K-3)])

fcast.ar4 # Print to console
```

```
##           [,1]
## [1,] 2.415979
```

```
# Forecast error
fcast.ar4.error <- as.numeric(GDPGrowth["2003"][1]) - fcast.ar4
c(fcast.ar1.error,fcast.ar4.error) # Print to console
```

```
## [1] -0.3858802 -0.3944846
```

One-Step Ahead US GDP Growth Forecast

$$Y_t = 2.05 + 0.26Y_{t-1} + 0.15Y_{t-2} + 0.04Y_{t-3} + 0.03Y_{t-4} + \hat{u}_t$$

(0.55) (0.09) (0.10) (0.08) (0.10)

- The one-step ahead forecast for US GDP growth is modeled as:

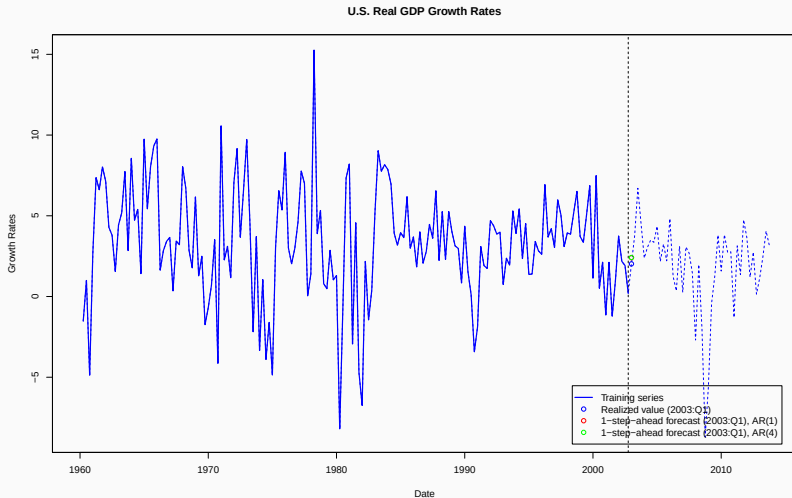
$$\hat{Y}_{T+1|T} = 2.05 + 0.26Y_T + 0.15Y_{T-1} + 0.04Y_{T-2} + 0.03Y_{T-3} + \hat{u}_T$$

(0.55) (0.09) (0.10) (0.08) (0.10)

- where the numbers in parentheses are the standard errors:
- The forecast error is:

$$Y_{T+1} - \hat{Y}_{T+1|T}$$

ONE-STEP AHEAD US GDP GROWTH FORECAST



Stationarity and AR(1)

So far, we just dealt with how to estimate AR models without thinking about its properties.

Unlike MA processes which are always stationary, AR processes require parameter restrictions for stationarity to hold.

If stationarity does not hold, above estimation and inference methods on β s require further modifications to be valid - we will come back to this in week 9.

Auto-Correlation of AR(1)

- AR(1): non-adjacent values are correlated, although the further apart they are the weaker the correlation. Unlike MA(p) processes where ACF cuts off after p lags, dependence in AR processes decay gradually/geometrically.
- For ease of derivation, we will look at simplified AR (1) without intercept below.

Auto-Correlation of AR(1)

Let's say process u_t (with zero mean) following AR(1) is built from the current and all past shocks (white noise). We replace past u as follows

$$\begin{aligned}u_t &= \rho u_{t-1} + \epsilon_t \\&= \rho(\rho u_{t-2} + \epsilon_{t-1}) + \epsilon_t \\&= \rho^2 u_{t-2} + \rho \epsilon_{t-1} + \epsilon_t \\&\vdots \\&= \rho^3 u_{t-3} + \rho^2 \epsilon_{t-2} + \rho \epsilon_{t-1} + \epsilon_t \\&\vdots\end{aligned}$$

Auto-Correlation of AR(1)

The auto-correlations for the AR(1) process are given by:

- $\text{Corr}(u_t, u_{t-1}) = \rho$
- $\text{Corr}(u_t, u_{t-2}) = \rho^2$

and so on.

AR(1): Stationarity

The AR(1) is stationary, only if we impose the following additional conditions:

- The process has been going for all time (starting at $-\infty$)
- And $|\rho| < 1$.

Summary

- Autoregressions predict the future using the immediate past
- The order of the autoregression is the number of periods considered in the immediate past
- Estimation of autoregressions by OLS
- Use estimated autoregression to forecast

Understanding Autoregressive (AR) Models

We have so far focused on **AR models**, mainly for **forecasting purposes**, not for causal analysis, where **lagged** Y_t such as Y_{t-1} and Y_{t-2} are used as regressors.

We relied on **OLS estimation**, but we need to carefully consider the **assumptions** required for **OLS estimates to be consistent**.

Extending to ADL Models

We will revisit this in the next lecture for an **extended model** that includes **lagged** Y_t as well as other regressors, known as **Autoregressive Distributed Lag Models (ADL)**.

Static Specification

Let's consider a **simple static model** (i.e., no lagged values on the right-hand side):

$$Y_t = \beta_0 + \beta_1 X_t + u_t$$

Simple Linear Regression

Consider the simple linear regression model:

$$Y_t = \beta_0 + \beta_1 X_t + u_t$$

Here, we assume the error term u_t exhibits **autocorrelation**, meaning the covariance between error terms at different time periods is not zero:

$$\text{Cov}(u_t, u_s) \neq 0$$

Impact on OLS Estimation

- **Standard Errors:** The standard errors of Ordinary Least Squares (**OLS**) estimators are affected by autocorrelation. They are no longer valid for inference in the usual way.
- **Unbiasedness and Consistency:** Importantly, autocorrelation **does not** impact the unbiasedness and consistency of **OLS** estimators. They remain unbiased and consistent.

Robust Inference

For valid statistical inference (hypothesis testing and confidence intervals) under autocorrelation, we must use:

ROBUST STANDARD ERRORS

This approach is analogous to using robust standard errors when dealing with heteroscedasticity.

ASSUMPTION

In order to establish the properties of regression models with time series data, we need to revise the standard assumptions to take into account the possible dependence across observations.

- **[C1]** The model is linear in the parameters and correctly specified
- **[C2]** The time series (Y_t, X_t) are weakly stationary and (Y_t, X_t) and (Y_{t-j}, X_{t-j}) become independent as j gets large (*weak dependence*)
- **[C3]** There is no exact linear relationship among regressors (*no perfect multicollinearity*)
- **[C4] Exogeneity:** $E(u_t | X_t, X_{t-1}, \dots) = 0$ for all t : conditioning set is current and past values of X
- **[C4-Strict] Strict Exogeneity:**
 $E(u_t | \dots, X_{t+1}, X_t, X_{t-1}, \dots) = 0$ for all t : conditional on past, current and **future** X

ASSUMPTIONS

$$y_t = \beta_0 + \beta_1 x_t + u_t.$$

- [C5] $\text{Var}(u_t) = \sigma^2$ for all t (**homoscedasticity**): will be relaxed
- [C6] $\text{Cov}(u_t, u_s) = 0$ for $s \neq t$: ie **no autocorrelation** in the error term: will be relaxed

Assumption C6 is commonly violated in Time Series: errors are usually correlated across time. Always use robust standard errors that relax both C5 and C6- we do not want to wrongly impose C5 and/or C6.

ASSUMPTIONS

$$y_t = \beta_0 + \beta_1 x_t + u_t.$$

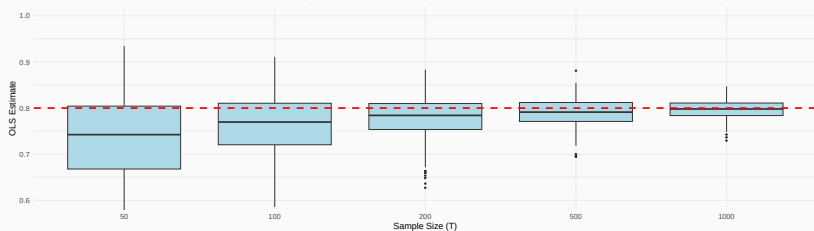
Under C1-C4-strict, the OLS of β_1 is **unbiased**. However Strict exogeneity is a strong condition that needs to be relaxed in many economic applications.

Under C1-C4, OLS is **consistent**, although not unbiased. For T large enough, we are happy with consistency

ASSUMPTIONS

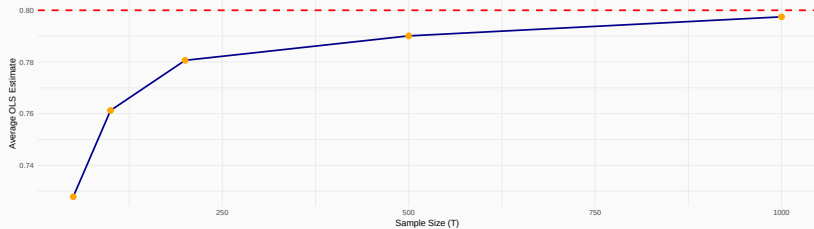
Distribution of OLS Estimates by Sample Size

Red dashed line : True $\beta_1 = 0.8$ for population regression $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$



Average OLS Estimate Converges to True Value

Consistency: Bias diminishes as T increases



DISTRIBUTED LAG MODELS

Suppose you have the **distributed lag (DL) model**:

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 x_{t-1} + \beta_3 x_{t-2} + u_t, \quad (2)$$

where x_t are **exogenous**.

We can estimate **dynamic causal effects** from the model above.

INTERPRETATION OF THE COEFFICIENTS WITH LAGGED x_t

- The estimate of β_1 represents the effect of x on y . What about the coefficients β_2 and β_3 ?
- β_2 : effect of unit change in X on Y 1 period later, representing dynamic causal effect.
- $\beta_1, \beta_2, \beta_3, \dots$: sequence of dynamic causal effects.

CONTEMPORANEOUS MULTIPLIER & DYNAMIC MULTIPLIER

- From (2), we see that the current effect of x on y , called the contemporaneous multiplier or impact effect, is given by $\hat{\beta}_1$.

$$\text{If } \Delta x_t = 1, \Delta y_t = \beta_1$$

- β_{h+1} is the h -period dynamic multiplier.

INTERPRETATION OF THE COEFFICIENTS WITH LAGGED x_t

- Assume that model (2) reaches “equilibrium’’, and denote by x^e and y^e the values of x and y in equilibrium (intuitively, equilibrium is reached when the values of x and y do not change anymore in time, i.e. they become stable)
- Model 2 in equilibrium is

$$\begin{aligned}y^e &= \beta_0 + \beta_1 x^e + \beta_2 x^e + \beta_3 x^e \\ &= \beta_0 + (\beta_1 + \beta_2 + \beta_3) x^e.\end{aligned}\tag{3}$$

INTERPRETATION OF THE COEFFICIENTS WITH LAGGED x_t

- Once estimates of β s are obtained from (2), we can use the equilibrium equation to deduce that the long-run cumulative dynamic multiplier of x on y is given by $\hat{\beta}_1 + \hat{\beta}_2 + \hat{\beta}_3$. If x of all periods increase by 1 unit, then y would increase by $\beta_1 + \beta_2 + \beta_3$

Possible problem of multicollinearity between X_t , X_{t-1} , X_{t-2} if X_t is correlated over time

SUMMARY

- Distributed Lag (**DL**) model can be used to estimate dynamic causal effects
- C1-C4 imply **OLS** is consistent
- Always standard errors robust to autocorrelation
- Next time, we combine **AR** and **DL** models to get **ADL** model